Foodie-Fi — SQL Case Study #3

8 Week SQL Challenge by Danny Ma

Section A: Customer Journey

A1. Based on the 8 sample customers provided in the sample from the subscriptions table, write a brief description about each customer's onboarding journey.

SQL Query:

```
SELECT
    customer_id,
    'Customer ' || customer_id ||
    ' started their subscription on ' || TO_CHAR(MIN(start_date), 'YYYY-MM-DD') ||
    ' and last subscribed on ' || TO_CHAR(MAX(start_date), 'YYYY-MM-DD') ||
    '. They tried ' || COUNT(DISTINCT plan_id) || ' plan(s), which are: ' ||
    STRING_AGG(DISTINCT plan_name::text, ', ') || '.' AS description
FROM subscriptions
JOIN plans USING (plan_id)
WHERE customer_id IN (1, 2, 11, 13, 15, 16, 18, 19)
GROUP BY customer_id
ORDER BY customer_id;
```

Output:

	customer_id integer	description text
1	1	Customer 1 started their subscription on 2020-08-01 and last subscribed on 2020-08-08. They tried 2 plan(s), which are: basic monthly, trial.
2	2	Customer 2 started their subscription on 2020-09-20 and last subscribed on 2020-09-27. They tried 2 plan(s), which are: pro annual, trial.
3	11	Customer 11 started their subscription on 2020-11-19 and last subscribed on 2020-11-26. They tried 2 plan(s), which are: churn, trial.
4	13	Customer 13 started their subscription on 2020-12-15 and last subscribed on 2021-03-29. They tried 3 plan(s), which are: basic monthly, pro monthl...
5	15	Customer 15 started their subscription on 2020-03-17 and last subscribed on 2020-04-29. They tried 3 plan(s), which are: churn, pro monthly, trial.
6	16	Customer 16 started their subscription on 2020-05-31 and last subscribed on 2020-10-21. They tried 3 plan(s), which are: basic monthly, pro annual, ...
7	18	Customer 18 started their subscription on 2020-07-06 and last subscribed on 2020-07-13. They tried 2 plan(s), which are: pro monthly, trial.
8	19	Customer 19 started their subscription on 2020-06-22 and last subscribed on 2020-08-29. They tried 3 plan(s), which are: pro annual, pro monthly, tri...

Section B: Data Analysis Questions

B1. How many customers has Foodie-Fi ever had?

SQL Query:

```
SELECT COUNT(DISTINCT customer_id) AS total_customers
FROM subscriptions;
```

Output:

	total_customers bigint
1	1000

B2. What is the monthly distribution of trial plan start_date values for our dataset — use the start of the month as the group by value?

SQL Query:

```
SELECT
    EXTRACT(YEAR FROM start_date) AS year,
    EXTRACT(MONTH FROM start_date) AS month,
    COUNT(plan_id) AS total_plans
FROM subscriptions
WHERE plan_id = 0
GROUP BY
    EXTRACT(YEAR FROM start_date),
    EXTRACT(MONTH FROM start_date)
ORDER BY year, month;
```

Output:

	year numeric	month numeric	total_plans bigint
1	2020	1	88
2	2020	2	68
3	2020	3	94
4	2020	4	81
5	2020	5	88
6	2020	6	79
7	2020	7	89
8	2020	8	88
9	2020	9	87
10	2020	10	79
11	2020	11	75
12	2020	12	84

B3. What plan start_date values occur after the year 2020 for our dataset? Show the breakdown by count of events for each plan_name.

SQL Query:

```
SELECT plan_name,
       COUNT(*) AS events
FROM subscriptions
JOIN plans USING (plan_id)
WHERE start_date >= '2020-01-01'
GROUP BY plan_name
ORDER BY events;
```

Output:

	plan_name character varying (13)	events bigint
1	pro annual	258
2	churn	307
3	pro monthly	539
4	basic monthly	546
5	trial	1000

B4. What is the customer count and percentage of customers who have churned rounded to 1 decimal place?

SQL Query:

```
SELECT
    COUNT(*) AS total_count,
    ROUND(
        COUNT(customer_id) * 100.0 / (SELECT COUNT(*) FROM subscriptions),
        1
    ) AS percentage
FROM subscriptions
WHERE plan_id = 4;
```

Output:

	total_count bigint	percentage numeric
1	307	11.6

B5. How many customers have churned straight after their initial free trial — what percentage is this rounded to the nearest whole number?

SQL Query:

```
WITH next_plan AS (
    SELECT
        customer_id,
        plan_id,
        LEAD(plan_id) OVER (
```

```
            PARTITION BY customer_id
            ORDER BY start_date
        ) AS lead_plan
    FROM subscriptions
)
SELECT
    COUNT(customer_id) AS total_customers,
    ROUND(
        COUNT(customer_id) * 100.0
        / (SELECT COUNT(DISTINCT customer_id) FROM subscriptions),
        0
    ) AS percentage
FROM next_plan
WHERE plan_id = 0
  AND lead_plan = 4;
```

Output:

	total_customers bigint	percentage numeric
1	92	9

B6. What is the number and percentage of customer plans after their initial free trial?

SQL Query:

```
WITH next_plan AS (
    SELECT
        customer_id,
        plan_id,
        LEAD(plan_id) OVER (
            PARTITION BY customer_id
            ORDER BY start_date
        ) AS lead_plan
    FROM subscriptions
)
SELECT
    lead_plan AS plan_after_trial,
    COUNT(DISTINCT customer_id) AS total_customers,
    ROUND(
        COUNT(DISTINCT customer_id) * 100.0
        / (SELECT COUNT(DISTINCT customer_id) FROM subscriptions),
        1
    ) AS percentage
FROM next_plan
WHERE plan_id = 0
  AND lead_plan IS NOT NULL
GROUP BY lead_plan
ORDER BY total_customers DESC;
```

Output:

	plan_after_trial integer	total_customers bigint	percentage numeric
1	1	546	54.6
2	2	325	32.5
3	4	92	9.2
4	3	37	3.7

B7. What is the customer count and percentage breakdown of all 5 plan_name values at 2020-12-31?

SQL Query:

```
SELECT
    plan_name,
    COUNT(DISTINCT customer_id),
    ROUND(
        COUNT(DISTINCT customer_id) * 100.0
        / (SELECT COUNT(*) FROM subscriptions),
        3
    ) AS percentage
FROM subscriptions
JOIN plans USING (plan_id)
WHERE start_date = '2020-12-31'
GROUP BY plan_name;
```

Output:

	plan_name character varying (13)	count bigint	percentage numeric
1	churn	1	0.038

B8. How many customers have upgraded to an annual plan in 2020?

SQL Query:

```
WITH next_plan AS (
    SELECT
        customer_id,
        plan_id,
        LEAD(plan_id) OVER (
            PARTITION BY customer_id
            ORDER BY start_date
        ) AS lead_plan
    FROM subscriptions
)
SELECT
```

```
    COUNT(customer_id) AS total_customers,
    ROUND(
        COUNT(customer_id) * 100.0
        / (SELECT COUNT(DISTINCT customer_id) FROM subscriptions),
        2
    ) AS percentage
FROM next_plan
WHERE lead_plan = 3;
```

Output:

	total_customers bigint	percentage numeric
1	258	25.80

B9. How many days on average does it take for a customer to upgrade to an annual plan from the day they join Foodie-Fi?

SQL Query:

```
WITH annual_plan AS (
    SELECT customer_id,
           MIN(start_date) AS annual_plan_buy
    FROM subscriptions
    WHERE plan_id = 3
    GROUP BY customer_id
    ORDER BY customer_id
),
first_plan AS (
    SELECT customer_id,
           MIN(start_date) AS plan_buy
    FROM subscriptions
    WHERE plan_id IN (0, 1, 2, 4)
    GROUP BY customer_id
)
SELECT
    ROUND(AVG(annual_plan_buy - plan_buy), 0) AS avg_duration
FROM annual_plan
JOIN first_plan USING (customer_id);
```

Output:

	avg_duration numeric
1	105

B11. How many customers downgraded from a pro monthly to a basic monthly plan in 2020?

SQL Query:

```
WITH lag_plan AS (
    SELECT
        customer_id,
        plan_id,
        LAG(plan_id) OVER (
            PARTITION BY customer_id
            ORDER BY start_date
        ) AS lag_plan
    FROM subscriptions
)
SELECT
    COUNT(customer_id) AS total_customers
FROM lag_plan
WHERE lag_plan = 2
  AND plan_id = 1;
```

Output:

	total_customers bigint
1	0